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X-ray scattering signatures of β-thalassemia

Omar S. Desouky [a], Wael M. Elshemey [b,*], Nabila S. Selim [a]

[a] Radiation Physics Department, National Center for Radiation Research and Technology (NCRRT), Egypt
[b] Biophysics Department, Faculty of Science, Cairo University, Egypt

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ABSTRACT

X-ray scattering from lyophilized proteins or protein-rich samples is characterized by the presence of two characteristic broad peaks at scattering angles equivalent to momentum transfer values of 0.27 and 0.6 nm^{-1}, respectively. These peaks arise from the interference of coherently scattered photons. Once the conformation of a protein is changed, these two peaks reflect such change with considerable sensitivity. The present work examines the possibility of characterizing the most common cause of hemolytic anaemia in Egypt and many Mediterranean countries; β-thalassemia, from its X-ray scattering profile. This disease emerges from a genetic defect causing reduced rate in the synthesis of one of the globin chains that make up hemoglobin. As a result, structurally abnormal hemoglobin molecules are formed. In order to detect such molecular disorder, hemoglobin samples of β-thalassemia patients are collected, lyophilized and measured using a conventional X-ray diffractometer. Results show significant differences in the X-ray scattering profiles of most of the diseased samples compared to control. The shape of the first scattering peak at 0.27 nm^{-1}, in addition to the relative intensity of the first to the second scattering peaks, provides the most reliable signs of abnormality in diseased samples. The results are interpreted and confirmed with the aid of Fourier Transform Infrared (FTIR) spectroscopy of normal and thalassemia samples.

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1. Introduction

Thalassemia is the most common inherited genetic disorder of blood. It is characterized by the absence or reduced production in globin chains, resulting in microcytic anaemia of varying degrees [15]. The absence or reduced production in the two β-chains of globin results in β-thalathemia. In Egypt, β-thalassemia is the most common cause of chronic hemolytic anaemia [50].

The present methods for the detection of β-thalassemia is based on the detection of the β-thalassemia carrier, Hemoglobin A2 (HbA2), a naturally occurring hemoglobin, expressed at low concentration (2%–3%) in normal individuals but with exceptionally high level in patients having β-thalassemia [15]. The detection of HbA2 in case of mild thalassemia requires highly sensitive techniques where the concentrations of HbA2 are only 3.3% and 3.7% of the total hemoglobin. Such a small elevation in the level of HbA2 may be caused by many factors other than β-thalassemia [47]. Current molecular methods for β-thalassemia carrier screening remain prohibitively expensive and inefficient, especially for population-based screening in densely populated developing countries. These methods include conventional PCR-based strategies such as temporal Temperature Gradient Gel Electrophoresis (TGGE), Denaturing Gradient Gel Electrophoresis (DGGE), Reverse Dot Blot Hybridization (RDBH), Amplification Refractory Mutation System (ARMS) and Direct Sequencing and Multiplex Mini-Sequencing [54]. It is thus important to investigate new screening techniques or techniques that are supportive to the currently available ones.

In the last decade, there has been growing interest in the useful applications of X-ray scattering in the biological and medical fields. The work of [40] showed that measured diffraction patterns for some plastics and many biological samples were characteristic as they result from the interference of photons coherently scattered from molecules in the medium. Water-rich samples such as liver, muscle and blood had diffraction patterns similar to that of water due to their high water content. The fact that photon scattering has become a new source of information in medicine and biology is highlighted in [62].

An early trial to characterize breast tissues and tissue-substitute materials using a polychromatic source, which considerably affected their results, is presented in [21]. They suggested the use of a monochromatic source to characterize tissues of similar water content.

One of the useful applications that has been introduced since then is the development of imaging techniques making use of the measurements of scattered photons [2,3,5,7,10,12–14,17,18,25,27,31,33,37,39,41,42,44,49,56,68,69,70].

* Corresponding author.
E-mail address: waelelshemey@yahoo.com (W.M. Elshemey).

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Other researchers made use of X-ray scattering in the characterization of bones [1,4,12,23,24,43,48,53,57,58,67].

One important application of X-ray scattering is introduced in [38]. They illustrated that the differentiation between normal and malignant breast tissues would be made possible based on the difference between the ratio of fat to soft tissue in both cases, which resulted in different and characteristic X-ray scattering signatures. This was followed by valuable contributions to the same research area by many authors [5,6,9,22,26,28–30,36,51,52,59–61,64].

X-ray scattering from lyophilized blood and biological samples resulted in two characteristic peaks at scattering angles equivalent to momentum transfer values of 0.27 and 0.60 nm^{-1}, respectively [16,19]. The shape and relative intensities of these peaks were characteristic for the investigated samples, the 1st peak at 0.27 nm^{-1} being sensitive to the molecular structure of proteins and protein-rich samples and was able to reflect molecular level changes in such samples. Moreover, it has been possible to characterize liver cirrhosis and Hepatocellular Carcinoma (HCC) from the X-ray profiles of lyophilized serum samples of patients [20].

The present work investigates the possibility of using the sensitivity of coherent X-ray scattering profiles from lyophilized hemoglobin as a complementary diagnostic tool in the detection of β-thalassemia. The work also includes measurements using Fourier Transform Infrared (FTIR) spectroscopy, which has recently shown promising results [45] and would provide supportive molecular insight.

2. Theoretical background

Low-angle scattering of low-energy X-ray photons (e.g. 8.047 keV) is dominated by the elastic (coherent) scattering process. The energy transferred to the struck electron is less compared to the binding energy of the electron. The atom is neither ionized nor excited and the recoil momentum is absorbed by the entire atom. Under these conditions, the wavelength of the radiation scattered by the bound electrons of an atom is essentially the same as that of the incident photon. There is a fixed phase relationship among the scattered radiations, which are thus capable of producing constructive interference. The differential coherent scattering cross-section per atom for unpolarized radiation can be expressed approximately as:

$$\frac{d_a\sigma_{coh}}{d\Omega} = \frac{r_o^2}{2}(1+\cos^2\theta)[F(x,Z)]^2 \tag{1}$$

where r_o is the classical electron radius, θ is the photon scattering angle and,

$$\frac{r_o^2}{2}(1+\cos^2\theta) \tag{2}$$

is the energy independent Thomson differential cross-section per electron, $(d_e\sigma_T/d\Omega)$. The variable $F(x, Z)$ is called the atomic form factor, which is the sum of electronic form factors and represents the ratio of the amplitude of the coherently scattered radiation by an entire atom to that of a single free electron. The square of this form factor is the probability that Z electrons of the atom take up a recoil momentum, (q), without absorbing any energy. The form factor depends on the combined variable $x = \{\sin(\theta/2)\}/\lambda$, where λ is the wavelength of the incident photon.

When considering a molecule, the molecular form factor is calculated by adding the squares of the individual atomic form factors, weighted by their respective atomic abundance n_i, where

$$F_m^2(x) = \sum_i n_i F^2(x,Z_i) \tag{3}$$

For compounds of unknown atomic abundances, the molecular form factor is given by

$$\frac{F_m^2(x)}{W} = \sum_i \frac{w_i}{M_i} F^2(x,Z_i) \tag{4}$$

where w_i is the fractional weight factor of element i, M_i is the atomic mass of element i and W is the molecular weight . Thus the coherent differential scattering cross-section for a molecule is given by

$$\frac{d_m\sigma_{coh}}{d\Omega} = \frac{r_o^2}{2}(1+\cos^2\theta)F_m^2(x) \tag{5}$$

Theoretical aspects of X-ray scattering from biological samples has been addressed by many authors [8,34,35,51,63,64].

3. Materials and methods

3.1. Collection and preparation of samples

3.1.1. Clinical samples

Samples of heparinized venous blood are collected from a total of ten patients diagnosed as β-thalassemic patients. All patients are transfusion-dependent and are under the supervision of medical professionals. Blood from thalassemic patients is collected just before the transfusion in a routine biochemical assessment. Blood from five healthy individuals is used as a normal control for this study. All individuals are aged between 6 and 10 years, with average hemoglobin concentration between 3.4 and 7.7 g/dl. Patients are under chelation therapy with desferoxamine (DFO).

3.1.2. Hemoglobin extraction

Hemolysates of washed, packed erythrocytes are prepared by a modification in the method [65]. The blood is centrifuged at 1500 rpm for 15 min at 4 °C, then the plasma is removed and the packed cells are washed with 5 volumes saline, centrifuged and the saline is removed. The procedure is repeated for two additional times, and the saline is removed after the final wash. The packed cells are lysed with 2 volumes of deionized water, then the mixture is centrifuged at 10,000 rpm for 30 min at 4 °C, and the supernatant is extracted. The extracted Hb is lyophilized using a freeze dryer (Edwards, UK) at −60 °C and 70 mbar.

3.2. X-ray scattering measurements

A Shimadzu powder X-ray diffractometer working in reflection geometry is used in the present work. The device, working at 40 KV and 30 mA, uses a Cu target to produce 8.047 KeV highly collimated X-ray beam. Scattering angles from 2° up to 32.5° are scanned in a step of 0.25°. Powder samples are smeared on a rough glass slide mounted vertically on a rotating sample holder. The rotation is in a $(\theta{-}2\theta)$ mode. Diffraction data are collected using a scintillation detector employing a sodium iodide crystal, graphite monochromator and is interfaced to computer.

3.3. Fourier transform infrared spectroscopy measurements

FTIR measurements are carried out using a single beam fourier transform infrared spectrometer, FT/IR-430, Jascow, Japan. This instrument is equipped with a ceramic IR source, KBr beam splitter and DLATGS IR detector. The measurement wave range for this instrument is from 7800 to 400 cm^{-1} and the *signal-to-noise* ratio at a resolution of 4 cm^{-1} is 104:1 at 2200 cm^{-1}. The FTIR spectra of the samples are obtained in the spectral range of

4000–400 cm^{-1} with a scanning speed of 2 mm/s and a resolution of 4 cm^{-1}. The number of scans is set to "Auto". Lyophilized hemoglobin samples are mixed with appropriate amount of KBr powder and gently ground in a special mortar for 30 s. The mixture is then pressed into a homogenous disk at a pressure of 5 tons/cm^{-2} using a hydraulic pressing system (Riken Power, Riken Seiki Co. Ltd. Japan). The infrared measurements are performed immediately after preparing the disks. Data are refined (using 7-point Savitzky-Golay smoothing), area is normalized and the second derivative spectra are calculated.

4. Results and discussion

X-ray scattering profiles of control hemoglobin and hemoglobin from patients with β-thalassemia (Th. series) are presented in Fig. 1(a–f) and Fig. 2(a–d). The graphs are normalized at the 2nd peak of scattering at 0.6 nm^{-1}. In each figure, a sample of individual patient is plotted together with control hemoglobin for the purpose of comparison. Due to the diversity of variations in the scattering profiles of the hemoglobin of patients, it would be meaningless to average the scattering from individual samples of

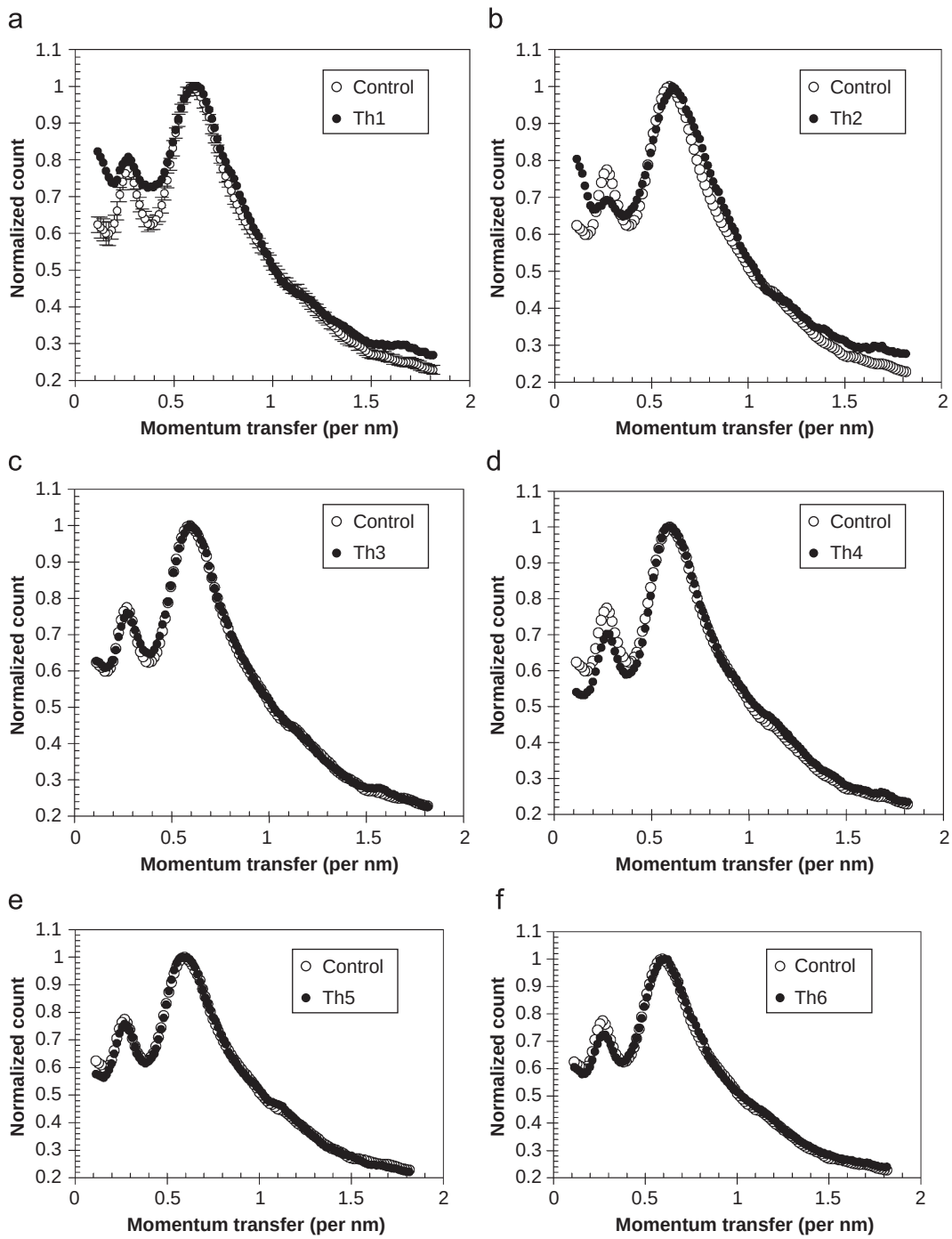

Fig. 1. (a–f). X-ray scattering from lyophilized hemoglobin of control and patients with β-thalassemia.

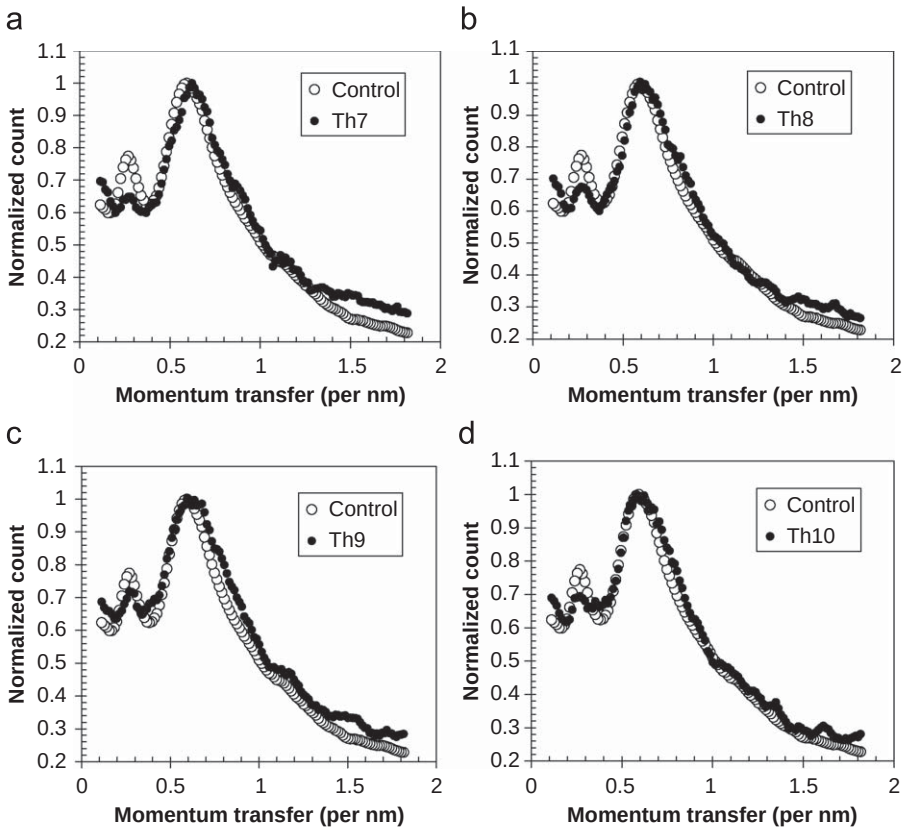

Fig. 2. (a–d) X-ray scattering from lyophilized hemoglobin of control and patients with β-thalassemia.

Table 1
Values of selected characterization parameters obtained from X-ray scattering profiles of control and β-Thalassemia patient's lyophilized hemoglobin.

	I_1/I_2 %	FWHM 1 (nm^{-1})	FWHM 2 (nm^{-1})	Peak 1 position (nm^{-1})	Peak 2 position (nm^{-1})
Control	77.44±1.7	0.094±0.009	0.26±0.004	0.27±0.002	0.59±0.003
Th.1	80.7	0.08	0.26	0.27	0.63
Th.2	68.86	0.07	0.31	0.27	0.6
Th.3	76	0.1	0.26	0.27	0.59
Th.4	70.2	0.1	0.28	0.27	0.59
Th.5	74.49	0.094	0.27	0.27	0.59
Th.6	71.26	0.1	0.27	0.27	0.61
Th.7	64.79	0.08	0.31	0.27	0.64
Th.8	66.83	0.08	0.3	0.27	0.62
Th.9	71.9	0.08	0.33	0.27	0.63
Th.10	68.83	0.08	0.3	0.27	0.62
Mean thalassemia	**71.386±3.5**	**0.0864±0.01**	**0.289±0.021**	**0.27**	**0.612±0.016**

patients in a single averaged profile. On the other hand, the diversity among control samples is small (represented by error bars in Fig. 1a) and thus an averaged profile for five control samples is used in Figs. 1 and 2. An overview over the scattered X-ray profiles of thalassemia patients reveals that the scattering profiles of most thalassemia samples show apparent differences compared to X-ray scattering profile of control hemoglobin. It is also apparent that the most prominent variations among patient's samples are the shape of the 1st scattering peak at 0.27 nm^{-1} and the ratio of intensities of the 1st and the 2nd scattering peaks (I_1/I_2%). Table 1 presents different parameters measured from the scattering profiles of control and thalassemia patients. Despite the presence of variations in the Full-Width at Half-Maximum (FWHM) of the 1st and the 2nd scattering peaks, and in the position of the 2nd scattering peak at 0.6 nm^{-1} of some samples compared to the control, yet these variations are subtle and may fall within standard error. In the majority of samples, the ratio I_1/I_2% is shown to be the most sensitive parameter towards the structural variations in hemoglobin of thalassemia patients compared to the control. For most samples, the ratio I_1/I_2% is found to decrease in thalassemia samples compared to the control. This observation agrees well with the observation of [20] on lyophilized serum scattering profiles of patients with liver cirrhosis and hepatocellular carcinoma. One may still wonder about the reason for the diversity of the shape of the 1st scattering peak among thalassemia patients and how would this be related to the molecular structural variations. One way of understanding this behavior may be the fact that β-thalassemia is extremely heterogeneous at the molecular level. Nearly 200 different mutations have been found to produce diverse defects in β-globin gene responsible for the β-globin chains in the hemoglobin of patients with β-thalassemia [11,15,66]. The last

two rows in Table 1 show the calculated mean values of different characterized parameters for thalassemia samples with their standard error in the last row.

The present work applies the well-established FTIR spectroscopy technique in the study of the control and some thalassemia lyophilized hemoglobin samples in order to reinforce the suggested molecular interpretations from X-ray scattering. For such purpose, only four thalassemia samples are thought to be quite conclusive. Fig. 3 presents area-normalized absorption FTIR spectrum for control and some β-thalassemia samples measured in the range of wave numbers from 4000 to 400 cm^{-1}. The modes most widely used in protein structural studies; amide I (1700–1600 cm^{-1}) and amide II (1600–1480 cm^{-1}) are well manifested. The amide I band arises principally from the C=O stretching vibration of the peptide group while the amide II band is primarily N–H bending with a contribution from C–N stretching vibrations. The amide I band of proteins consists of many overlapping component bands that represent different structural elements such as α-helices, β-sheets, turns and non-ordered or irregular structures [32]. Fig. 3 shows an apparent reduction in the area under peak in the amide I and amid II regions of thalassemia compared to the control. These regions are further magnified in Fig. 4, which in accordance with X-ray scattering measurements show a pronounced degree of diversity in the amount of reduction among different thalassemia samples. Fig. 5 presents the difference spectrum for thalassemia samples in the amide I and amide II regions obtained by subtraction of the area-normalized FTIR spectrum of individual samples from the control. The greater the reduction in area of a thalassemia sample compared to the control, the greater is the amplitude of its difference spectrum (e.g. the area of the Th1 sample in Fig. 4 shows maximum reduction compared to control and consequently its difference spectrum presented in Fig. 5 shows maximum amplitude compared to the other thalassemia samples). The reduction in amide I and amide II protein absorption bands provides strong evidence on the reduction of the amount of molecular components maintaining normal protein secondary structure. In other words, it offers solid indication about the presence of a degree of disturbance in the secondary structure of hemoglobin obtained from patients with β-thalassemia compared to the control [46,55]. Since the amide I band of protein consists of many overlapping bands corresponding to specific protein secondary structure elements, applying a band narrowing technique would possibly unveil the specific secondary structural changes accompanying β-thalassemia. According to [32,45], this would be achieved by plotting the second derivative of the FTIR spectrum. Fig. 6(a–d) presents the second derivative spectrum for the investigated control and β-thalassemia hemoglobin

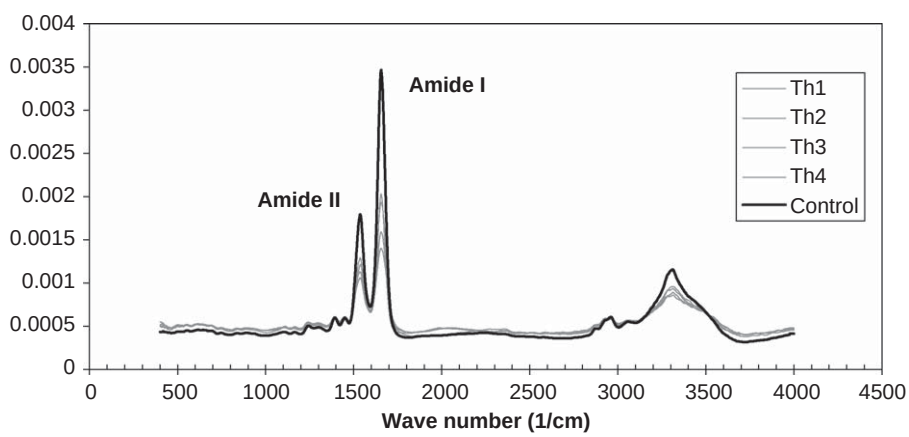

Fig. 3. Area-normalized FTIR spectra of some thalassemia samples and control hemoglobin.

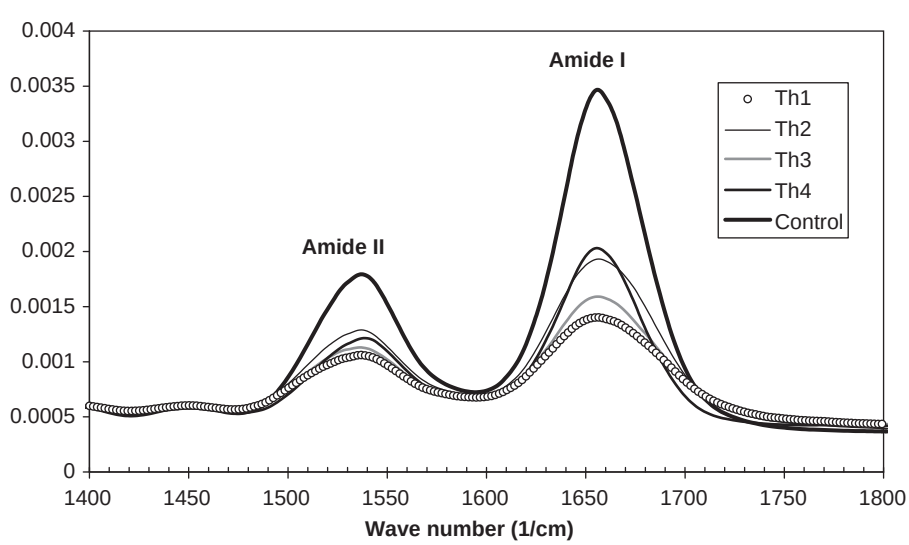

Fig. 4. Area-normalized FTIR spectra for thalassemic and control hemoglobin in the Amide I & II regions.

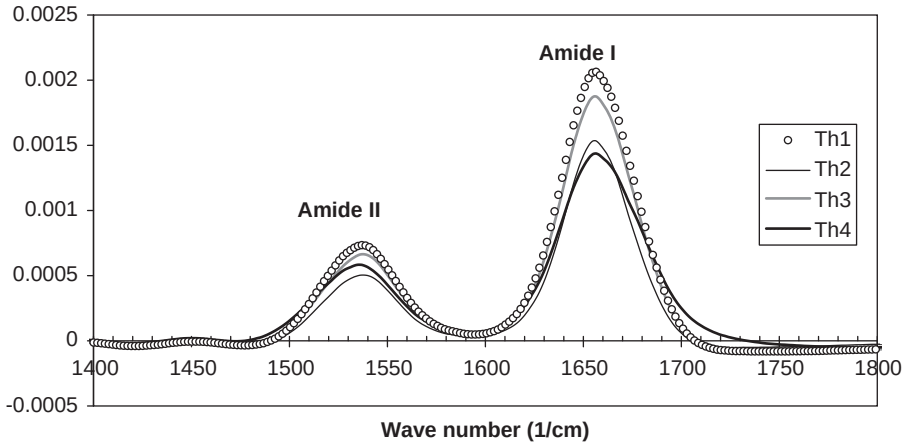

Fig. 5. Difference FTIR spectra for thalassemia samples from control in the Amide I & II regions.

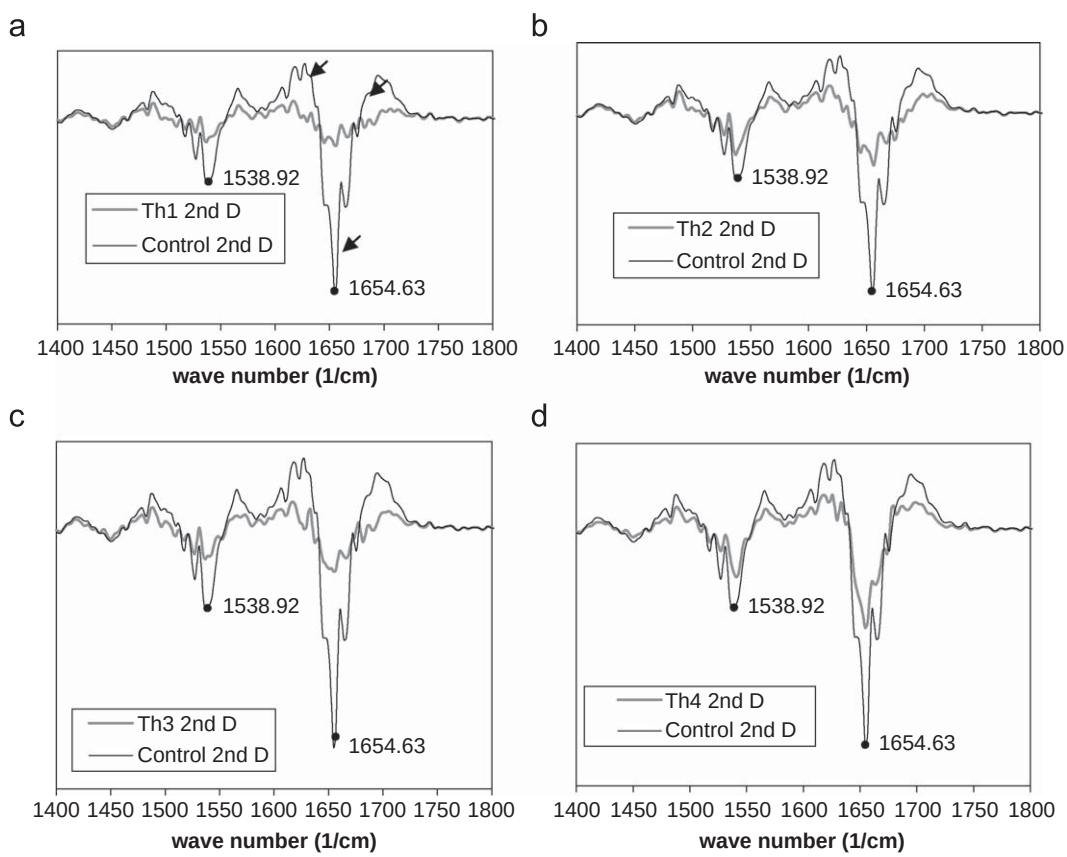

Fig. 6. (a–d) Second derivative spectra for control and some thalassemic hemoglobin samples.

samples. All thalassemia samples show an apparent reduction in the absorption peak at 1654 cm^{-1} indicating a sharp decrease in the α-helix content compared to the control. This decrease is accompanied by an increase in the low and high wave number regions corresponding to β-sheet content of thalassemia samples (around 1630 and 1680 cm^{-1}, respectively). The arrows in Fig. 6a point towards the α-helix peak and β-sheet regions showing maximum spectral deviations of β-thalassemia hemoglobin samples from the control. The amide II peak at 1538 cm^{-1} keeps on exhibiting a reduction in its intensity in case of thalassemia samples as previously mentioned in this paper. The current observations highly agree with the work of [45].

Based on the presented FTIR data, it would be reasonable to say that the reduction in the I_1/I_2% ratio of the X-ray scattering profiles of patients with β-thalassemia is probably due to a corresponding reduction in the α-helix content of patient samples compared to the control. This remark is supported by the work of [16], which shows that the X-ray scattering profile of Casein (a protein having

no well-defined secondary structure) has the 1st scattering peak at 0.27 nm^{-1} completely absent. Moreover, subjecting hemoglobin to heavy doses of radiation which would cause potential damage to the protein secondary structure has also shown to produce reduced I_1/I_2% ratio [19].

5. Conclusion

X-ray scattering from lyophilized hemoglobin of patients with β-thalassemia has shown characteristic features that would be invaluable in supporting current β-thalassemia screening techniques. It has been possible to obtain important molecular level interpretations of X-ray scattering profiles from patients with β-thalassemia through the analysis of FTIR spectroscopy data. FTIR technique, itself, has also proven to be very sensitive in the characterization of β-thalassemia. Both techniques would probably present a promising approach in the screening of β-thalassemia. In future work, a survey involving larger number of individuals and extensive statistical analysis is expected to enhance the understanding of the uncertainty in the measurements and to improve its accuracy in order to establish the use of both techniques in the screening of β-thalassemia.

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